

Table 1: Calibration Quality Averaged Over Tree Models and Horizons. All Calibrators Are Fitted on Cross-Fitted Out-of-Fold Scores. Slope 1 / Intercept 0 Indicate Perfect Probabilistic Calibration.

Dataset	Method	AUC (fold mean)	Brier	ECE	Log loss	Slope	Intercept
NASA	Isotonic	0.854	0.207	0.151	0.652	0.48	-0.26
NASA	Platt	0.908	0.196	0.136	0.575	0.90	-0.14
NASA	Temperature	0.908	0.178	0.107	0.536	1.22	-0.30
CALCE	Isotonic	0.915	0.067	0.049	0.279	0.41	1.17
CALCE	Platt	0.946	0.073	0.029	0.280	0.01	2.54
CALCE	Temperature	0.945	0.080	0.067	0.284	0.53	1.42

Supplementary Material for: Multi-Horizon Failure-Risk Models for Battery Failure Prediction: How Much Cross-Chemistry Transfer Is a State-of-Health Shortcut?

Anonymous

This supplement collects the tables and figures omitted from the main text for length. All numbers are produced by the same pipeline as the main text.

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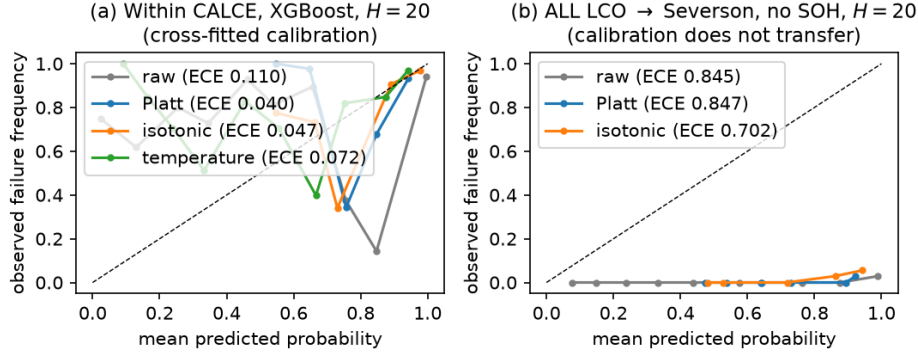


Figure 1: Reliability diagrams under the cross-fitted protocol. (a) Within CALCE, XGBoost at $H=20$: the fitted Platt sigmoid preserves ranking but washes confidence toward the base rate. (b) ALL-LCO → Severson without SOH at $H=20$: no fitted calibrator, whichever source scores it is applied to, repairs calibration on the shifted target.

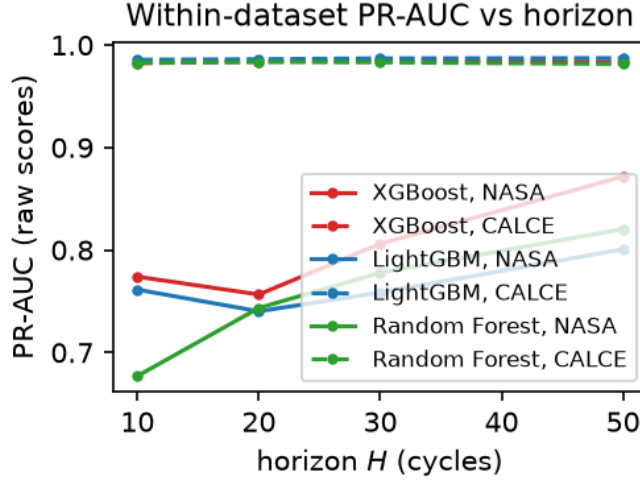


Figure 2: Within-dataset PR-AUC of raw scores versus horizon. Precision–recall performance, unlike ROC-AUC, is sensitive to the horizon-dependent failure prevalence; it is also where isotonic’s tied step outputs pay for their ranking-free compression.

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Table 2: GRU Transfer at H=20 on the Common Feature Set: Seed-Averaged Per-Cell AUC ($\pm$ SD Over Seeds 42, 1, 7).

Training	Feature set	Oxford	Severson
NASA	common w/ SOH	0.196 $\pm$ 0.155	0.998 $\pm$ 0.003
NASA	common no SOH	0.147 $\pm$ 0.210	0.987 $\pm$ 0.015
CALCE	common w/ SOH	0.182 $\pm$ 0.311	0.964 $\pm$ 0.033
CALCE	common no SOH	0.099 $\pm$ 0.171	0.479 $\pm$ 0.451
ALL LCO	common w/ SOH	0.088 $\pm$ 0.084	0.991 $\pm$ 0.015
ALL LCO	common no SOH	0.000 $\pm$ 0.000	0.893 $\pm$ 0.131

Table 3: Minimal Transfer Baselines at H=20: Pooled AUC With Cell-Level Bootstrap 95% CI. The Distance Rule Is the Unfitted Score 0.80–SOH. [Oxford target].

Baseline	NASA	CALCE	ALL
SOH distance rule	1.000 [1.000,1.000]	1.000 [1.000,1.000]	1.000 [1.000,1.000]
SOH only	1.000 [1.000,1.000]	1.000 [1.000,1.000]	1.000 [1.000,1.000]
cycle only	0.989 [0.983,1.000]	0.756 [0.732,0.780]	0.787 [0.760,0.814]
SOH + cycle	0.990 [0.985,1.000]	0.797 [0.770,0.825]	0.818 [0.788,0.848]
sensors only	0.500 [0.500,0.500]	0.500 [0.500,0.500]	0.500 [0.500,0.500]
full (7 feats)	0.500 [0.500,0.500]	0.500 [0.500,0.500]	0.500 [0.500,0.500]

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Table 4: Minimal Transfer Baselines at H=20: Pooled AUC With Cell-Level Bootstrap 95% CI. The Distance Rule Is the Unfitted Score 0.80–SOH. [Severson target].

Baseline	NASA	CALCE	ALL
SOH distance rule	0.912 [0.836,0.998]	0.912 [0.836,0.998]	0.912 [0.836,0.998]
SOH only	0.912 [0.836,0.998]	0.912 [0.836,0.998]	0.912 [0.836,0.998]
cycle only	0.713 [0.591,0.798]	0.713 [0.591,0.798]	0.713 [0.591,0.798]
SOH + cycle	0.865 [0.834,0.902]	0.764 [0.674,0.829]	0.758 [0.664,0.825]
sensors only	0.301 [0.015,0.549]	0.498 [0.497,0.500]	0.303 [0.059,0.497]
full (7 feats)	0.823 [0.675,0.995]	0.712 [0.589,0.797]	0.674 [0.525,0.782]

Table 5: Factorial Feature Ablation at H=20 (XGBoost, Pooled AUC With Cell-Level Bootstrap 95% CI). Feature Groups: SOH; Cycle Index; Sensors [Oxford target].

Feature set	NASA	CALCE	ALL
Full	0.957 [0.931,0.987]	0.850 [0.764,0.910]	0.917 [0.878,0.965]
No SOH	0.518 [0.513,0.522]	0.416 [0.264,0.554]	0.429 [0.279,0.597]
No cycle index	0.965 [0.942,0.989]	0.967 [0.929,0.981]	0.993 [0.954,1.000]
No SOH, no cycle	0.658 [0.522,0.797]	0.433 [0.282,0.594]	0.388 [0.224,0.506]
SOH only	0.965 [0.942,0.989]	1.000 [1.000,1.000]	1.000 [1.000,1.000]
Cycle only	0.541 [0.537,0.545]	0.541 [0.537,0.545]	0.541 [0.537,0.545]
Sensors only	0.658 [0.522,0.797]	0.433 [0.282,0.594]	0.388 [0.224,0.506]

Table 6: Factorial Feature Ablation at H=20 (XGBoost, Pooled AUC With Cell-Level Bootstrap 95% CI). Feature Groups: SOH; Cycle Index; Sensors [Severson target].

Feature set	NASA	CALCE	ALL
Full	0.871 [0.767,0.997]	0.850 [0.749,0.967]	0.896 [0.812,0.995]
No SOH	0.595 [0.510,0.726]	0.682 [0.670,0.695]	0.750 [0.692,0.835]
No cycle index	0.854 [0.733,0.998]	0.766 [0.584,0.974]	0.774 [0.578,0.998]
No SOH, no cycle	0.531 [0.433,0.680]	0.500 [0.500,0.500]	0.535 [0.455,0.653]
SOH only	0.834 [0.695,0.991]	0.798 [0.638,0.978]	0.833 [0.689,0.993]
Cycle only	0.709 [0.695,0.724]	0.694 [0.681,0.708]	0.719 [0.704,0.735]
Sensors only	0.531 [0.433,0.680]	0.500 [0.500,0.500]	0.535 [0.455,0.653]

Table 7: Same-Chemistry Cross-Laboratory Controls at $H=20$ (XGBoost, Pooled AUC [CI]). Full Feature Sets Exercise the Zero-Imputation Confound; Common Sets Restrict Both Datasets to the Shared Four-Feature Subset.

Pair	with SOH (full)	no SOH (full)	no SOH (common)
NASA→CALCE (LCO→LCO)	0.870 [0.799,0.966]	0.849 [0.753,0.985]	0.781 [0.708,0.964]
CALCE→NASA (LCO→LCO)	0.543 [0.356,0.677]	0.566 [0.427,0.678]	0.601 [0.496,0.703]
Severson→Oxford (LFP→LFP)	0.997 [0.991,1.000]	0.471 [0.466,0.477]	0.428 [0.422,0.435]
Oxford→Severson (LFP→LFP)	0.688 [0.605,0.788]	0.500 [0.500,0.500]	0.500 [0.500,0.500]

Table 8: Discrete-Time Hazard Model at $H=20$: Transfer Performance (Pooled AUC [CI]). The Fixed-Horizon XGBoost Comparator With SOH Is in main-text Table 3.

Training	Model	Oxford	Severson
NASA	Hazard XGBoost	0.971 [0.953,0.991]	0.884 [0.786,0.998]
NASA	Hazard Logistic	0.500 [0.500,0.500]	0.607 [0.402,0.890]
CALCE	Hazard XGBoost	0.466 [0.301,0.578]	0.737 [0.711,0.774]
CALCE	Hazard Logistic	0.500 [0.500,0.500]	0.500 [0.500,0.500]
ALL LCO	Hazard XGBoost	0.314 [0.135,0.457]	0.866 [0.770,0.978]
ALL LCO	Hazard Logistic	0.500 [0.500,0.500]	0.916 [0.873,0.969]

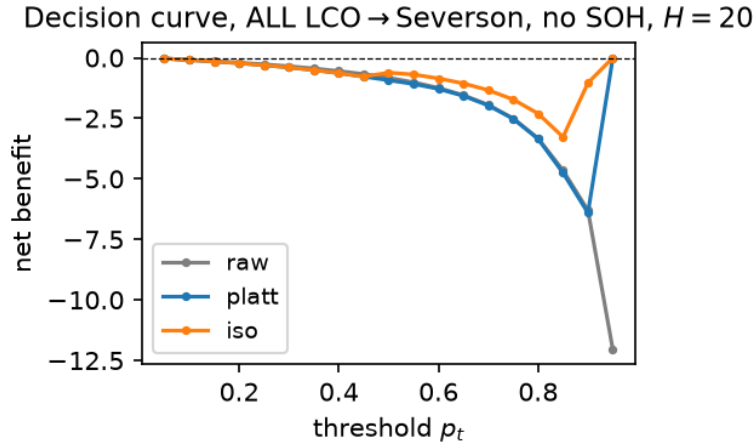


Figure 3: Decision-curve analysis (net benefit vs. threshold) for ALL-LCO → Severson without SOH at $H=20$: no score type produces a usable operating point once SOH is removed.

Table 9: Monotonicity Violations: Share of Scored Rows With $p_{10} \leq p_{20} \leq p_{30} \leq p_{50}$ Failing, on Raw Scores (Platt in Parentheses). The Survival Product Is Monotone by Construction (Rate 0).

Setting	Target	Model	Raw	Platt
within	calce	XGBoost	0.452	0.056
within	nasa	XGBoost	0.323	0.260
transfer (calce src)	oxford	LightGBM	0.641	0.004
transfer (calce src)	oxford	Random Forest	0.501	0.540
transfer (calce src)	oxford	XGBoost	0.565	0.003
transfer (nasa+calce src)	oxford	LightGBM	0.210	0.648
transfer (nasa+calce src)	oxford	Random Forest	0.335	0.305
transfer (nasa+calce src)	oxford	XGBoost	0.609	0.341
transfer (nasa src)	oxford	LightGBM	0.521	0.468
transfer (nasa src)	oxford	Random Forest	0.559	0.376
transfer (nasa src)	oxford	XGBoost	0.248	0.124
transfer (calce src)	severson	LightGBM	0.405	0.013
transfer (calce src)	severson	XGBoost	0.420	0.072
transfer (nasa src)	severson	LightGBM	0.582	0.537
transfer (nasa src)	severson	Random Forest	0.778	0.557
transfer (nasa src)	severson	XGBoost	0.958	0.923
—	—	Survival product	0.000	0.000

Table 10: Arm A Recalibration on Severson Without SOH ($H=20$): Values Average the Three LCO Sources and Available Tree Models; ECE Uses 10 Equal-Width Probability Bins. From the Released Recalibration Study Artifacts.

k	Zero ECE	Iso. ECE	Platt ECE	Temp. ECE	Iso. AUC	Platt AUC	Temp. AUC
5	0.817	0.044	0.045	0.563	0.616	0.599	0.669
10	0.817	0.010	0.010	0.561	0.634	0.646	0.651
20	0.817	0.013	0.013	0.560	0.634	0.655	0.648
40	0.817	0.015	0.015	0.561	0.637	0.657	0.653

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Table 11: Arm B Recovery at $k = 5$ on Severson Without SOH, $H = 20$. Recovery Ratio Uses the Within-LCO Ceiling; Retention Is the LCO-Holdout AUC Before and After the Update. *Ratio above 1.0 is a ceiling-mismatch artifact, not super-ceiling recovery (see text).

Training	Model	Zero AUC	Arm B AUC	Recovery	DeLong p	LCO retention
NASA	XGBoost	0.620	0.854	+1.04*	$< 10^{-3}$	1.000 $\rightarrow$ 1.000
NASA	LightGBM	0.493	0.838	+0.88	$< 10^{-3}$	1.000 $\rightarrow$ 0.997
NASA	Random Forest	0.724	0.751	+0.19	$< 10^{-3}$	0.959 $\rightarrow$ 0.314
CALCE	XGBoost	0.686	0.796	+0.53	$< 10^{-3}$	0.997 $\rightarrow$ 0.996
ALL LCO	XGBoost	0.760	0.828	+0.47	$< 10^{-3}$	0.690 $\rightarrow$ 0.567

Table 12: SOH Ablation Evidence (ALL-LCO Source): Pooled AUC With and Without SOH, Paired Cell-Level Bootstrap 95% CI on the Difference (Primary), and Cycle-Level DeLong p -Value (Secondary; Inflated by Within-Cell Correlation).

Target	Model	H	AUC w/ SOH	AUC w/o	Delta AUC [boot 95% CI]	DeLong p
Oxford	XGBoost	10	0.980	0.433	+0.547 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	XGBoost	20	0.917	0.429	+0.488 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	XGBoost	30	0.991	0.377	+0.614 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	XGBoost	50	0.932	0.580	+0.351 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	LightGBM	10	0.996	0.547	+0.449 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	LightGBM	20	0.971	0.332	+0.639 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	LightGBM	30	0.743	0.484	+0.259 (CI undefined: 5 cells)	$9.7e - 13$
Oxford	LightGBM	50	0.832	0.448	+0.384 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	10	0.978	0.626	+0.352 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	20	0.989	0.581	+0.409 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	30	0.971	0.462	+0.508 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	50	0.933	0.460	+0.473 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Severson	XGBoost	10	0.878	0.741	+0.137 [0.079, 0.249]	$< 10^{-16}$ (underflow)
Severson	XGBoost	20	0.896	0.750	+0.146 [0.097, 0.233]	$< 10^{-16}$ (underflow)
Severson	XGBoost	30	0.906	0.745	+0.162 [0.115, 0.237]	$< 10^{-16}$ (underflow)
Severson	XGBoost	50	0.917	0.736	+0.180 [0.134, 0.250]	$< 10^{-16}$ (underflow)
Severson	LightGBM	10	0.870	0.736	+0.134 [0.105, 0.200]	$< 10^{-16}$ (underflow)
Severson	LightGBM	20	0.882	0.718	+0.164 [0.127, 0.227]	$< 10^{-16}$ (underflow)
Severson	LightGBM	30	0.872	0.771	+0.102 [0.030, 0.230]	$< 10^{-16}$ (underflow)
Severson	LightGBM	50	0.899	0.744	+0.155 [0.112, 0.214]	$< 10^{-16}$ (underflow)
Severson	Random Forest	10	0.903	0.740	+0.163 [0.133, 0.225]	$< 10^{-16}$ (underflow)
Severson	Random Forest	20	0.889	0.746	+0.143 [0.106, 0.226]	$< 10^{-16}$ (underflow)
Severson	Random Forest	30	0.919	0.748	+0.171 [0.138, 0.232]	$< 10^{-16}$ (underflow)
Severson	Random Forest	50	0.921	0.746	+0.175 [0.114, 0.264]	$< 10^{-16}$ (underflow)

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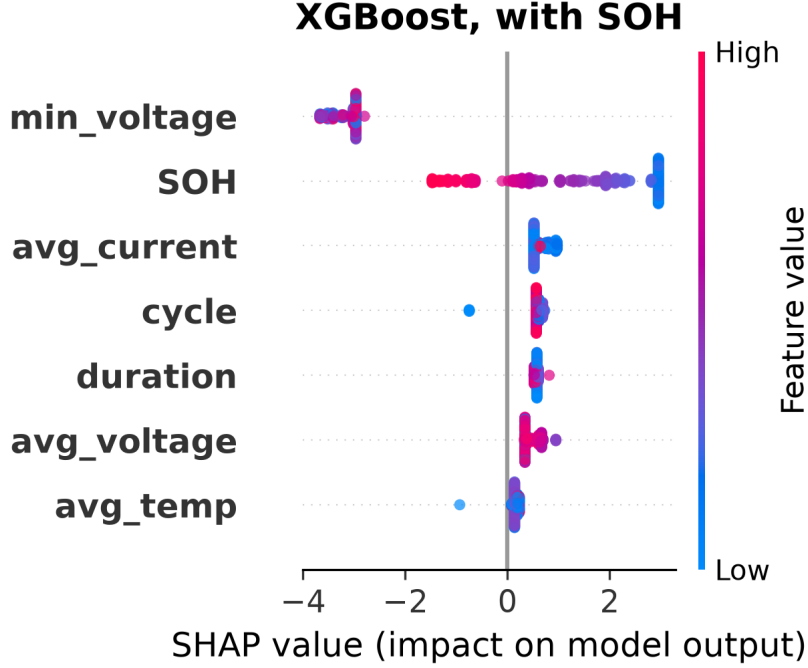


Figure 4: SHAP feature importance for XGBoost in NASA-to-Oxford transfer at H=20, with SOH. SOH dominates by a wide margin.

Table 13: Embedded Deployment: Model Size, Accuracy, On-Device Validation Error, and Measured Inference Time. Errors Are Maximum $|C \text{ engine} - \text{predict_proba()}|$ Over All 1 028 Rows. Timings Measured On-Device at 240 MHz With the Binary in Flash (PROGMEM), PSRAM, or a Single Model in Internal SRAM.

Model	Nodes	Max Err	Flash	PSRAM	SRAM (1 model)
XGBoost	3 272	2.4×10^{-7}	0.76 ms	0.50 ms	0.32 ms
LightGBM	7 000	1.1×10^{-6}	1.55 ms	0.93 ms	0.43 ms
Random Forest	16 064	3.0×10^{-8}	2.36 ms	1.29 ms	0.50 ms
All three	26 336	—	4.7 ms	2.7 ms	—

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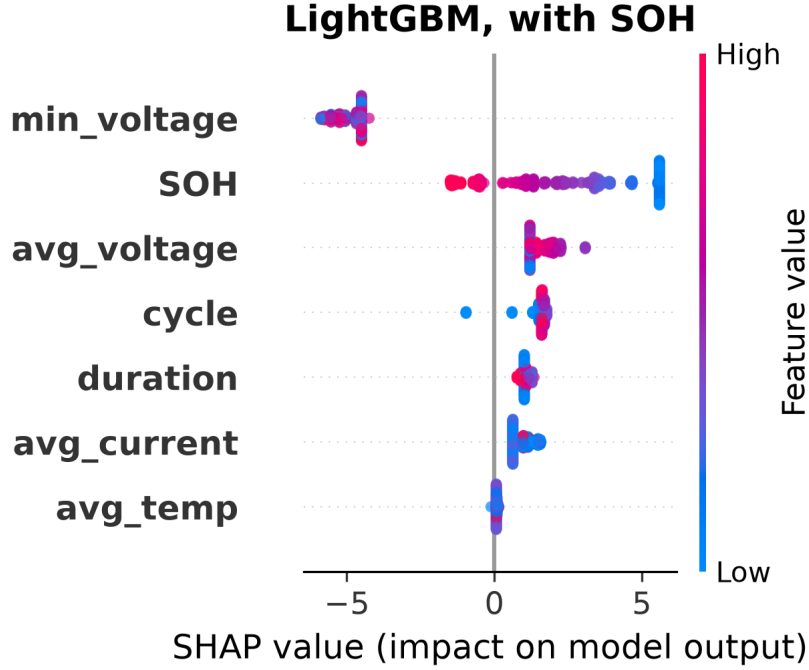


Figure 5: SHAP feature importance for LightGBM in NASA-to-Oxford transfer at $H=20$, with SOH. SOH dominates by a wide margin.

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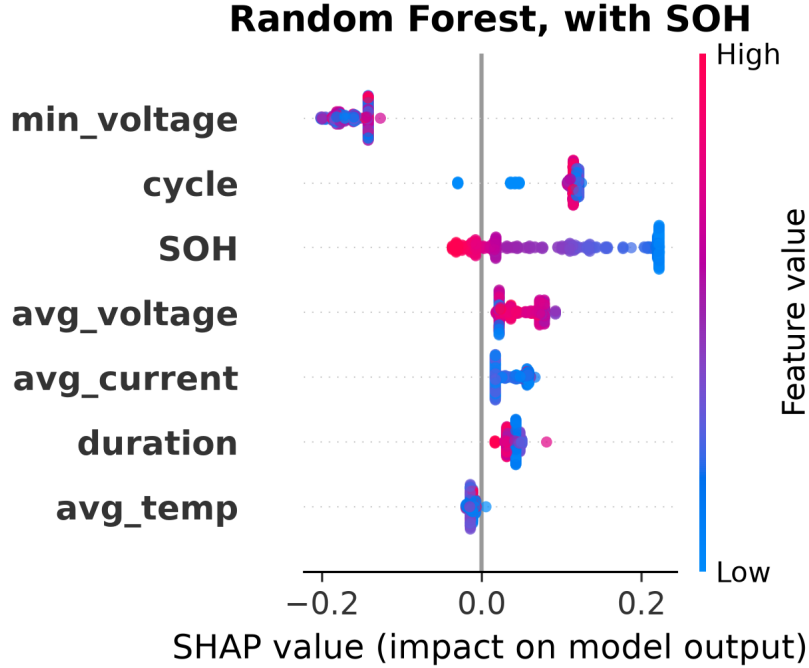


Figure 6: SHAP feature importance for Random Forest in NASA-to-Oxford transfer at H=20, with SOH. SOH dominates by a wide margin.

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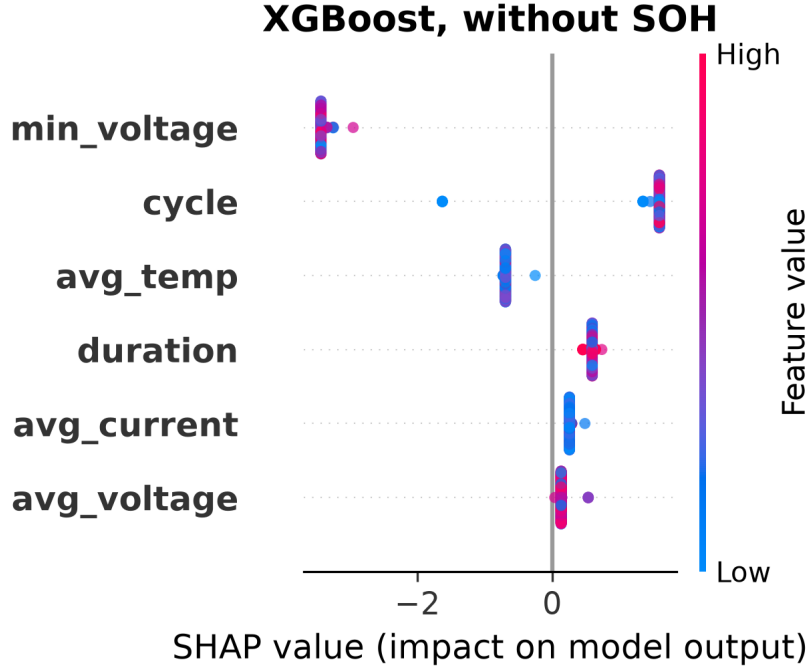


Figure 7: SHAP feature importance for XGBoost in NASA-to-Oxford transfer at H=20, without SOH. All features collapse to near-zero spread, so no transferable signal remains.

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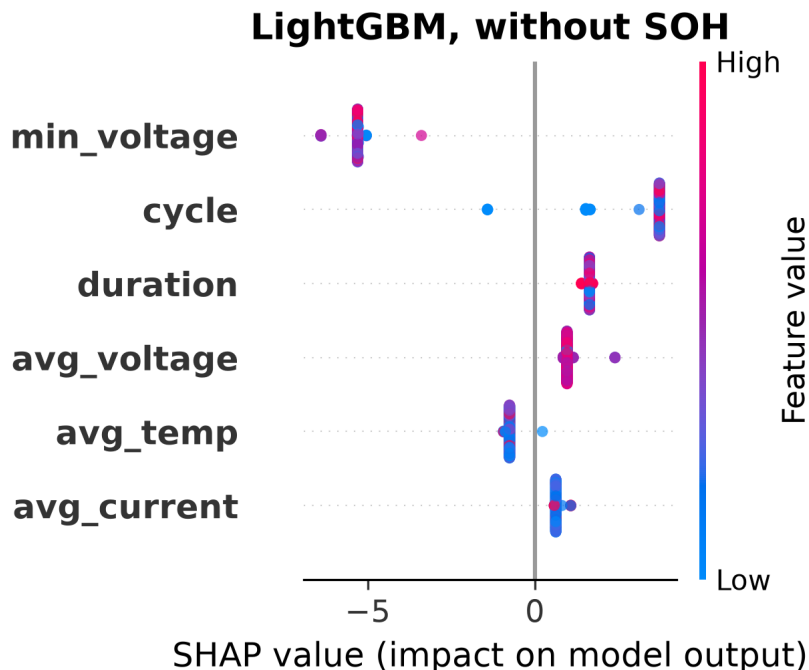


Figure 8: SHAP feature importance for LightGBM in NASA-to-Oxford transfer at H=20, without SOH. All features collapse to near-zero spread.

Table 14: Transfer to New Chemistries at H=20 (NASA+CALCE Source). Pooled AUC With Cell-Level Bootstrap 95% CI; Δ Is the Paired Per-Cell Bootstrap Difference (With – Without SOH). New Groups Are Transfer Targets Only.

Target	Model	With SOH	Without SOH	Delta (paired)
HNEI NMC	XGBoost	0.993 [0.983,0.999]	0.989 [0.977,0.997]	+0.004
HNEI NMC	LightGBM	0.999 [0.998,0.999]	0.988 [0.974,0.996]	+0.011
HNEI NMC	Random Forest	0.999 [0.999,1.000]	0.996 [0.994,0.999]	+0.003
SNL NCA	XGBoost	0.972 [0.956,0.988]	0.878 [0.824,0.930]	+0.094
SNL NCA	LightGBM	0.985 [0.974,0.993]	0.850 [0.791,0.911]	+0.135
SNL NCA	Random Forest	0.983 [0.974,0.992]	0.911 [0.875,0.951]	+0.072
SNL NMC	XGBoost	0.772 [0.677,0.979]	0.709 [0.612,0.953]	+0.063
SNL NMC	LightGBM	0.847 [0.743,0.982]	0.693 [0.570,0.959]	+0.153
SNL NMC	Random Forest	0.858 [0.748,0.975]	0.742 [0.689,0.874]	+0.115
SNL LFP	XGBoost	0.998 [0.997,1.000]	0.496 [0.458,0.531]	+0.502
SNL LFP	LightGBM	0.999 [0.997,1.000]	0.359 [0.279,0.443]	+0.640
SNL LFP	Random Forest	0.999 [0.998,1.000]	0.666 [0.569,0.769]	+0.332

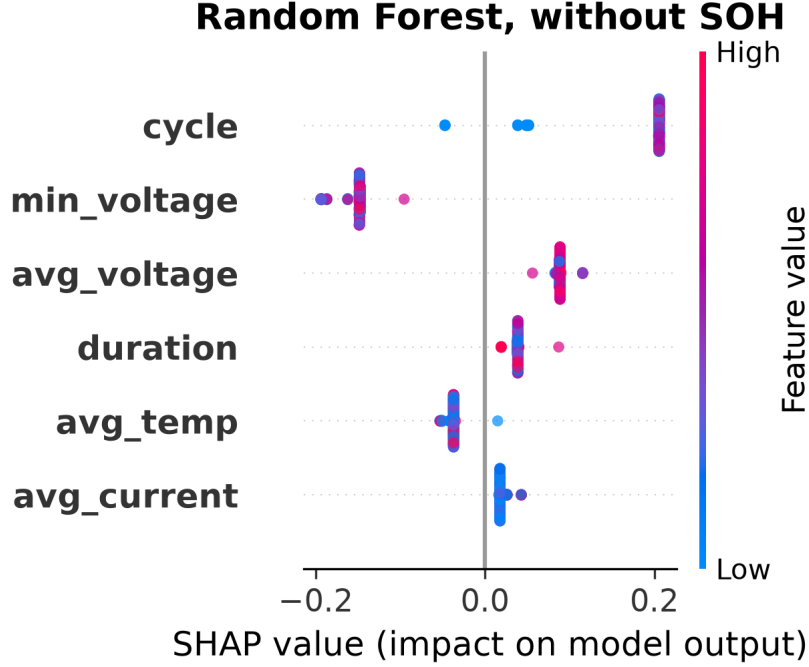


Figure 9: SHAP feature importance for Random Forest in NASA-to-Oxford transfer at H=20, without SOH. All features collapse to near-zero spread.

Table 15: Leave-Condition-Out Versus Mixed-Cell Folds at H=20 (XGBoost). Conditions Are Temperature Band, C-Rate, or Both; Metadata Never Enters the Features. Ref. Is the Ordinary Mixed-Cell AUC for the Same Feature Set (— Where Unstable or Not Run).

Group	Held-out axis	With SOH	Ref.	No SOH	Ref.
SNL NCA	temp $\times$ rate	0.989	0.985	0.894	0.928
SNL NCA	temperature	0.989	0.985	0.888	0.928
SNL NCA	C-rate	0.961	0.985	0.820	0.928
SNL NMC	temp $\times$ rate	0.847	—	0.666	—
SNL NMC	temperature	0.787	—	0.569	—
SNL NMC	C-rate	0.847	—	0.666	—
SNL LFP	temp $\times$ rate	0.991	0.994	—	—
SNL LFP	temperature	0.995	0.994	—	—
SNL LFP	C-rate	0.994	0.994	—	—

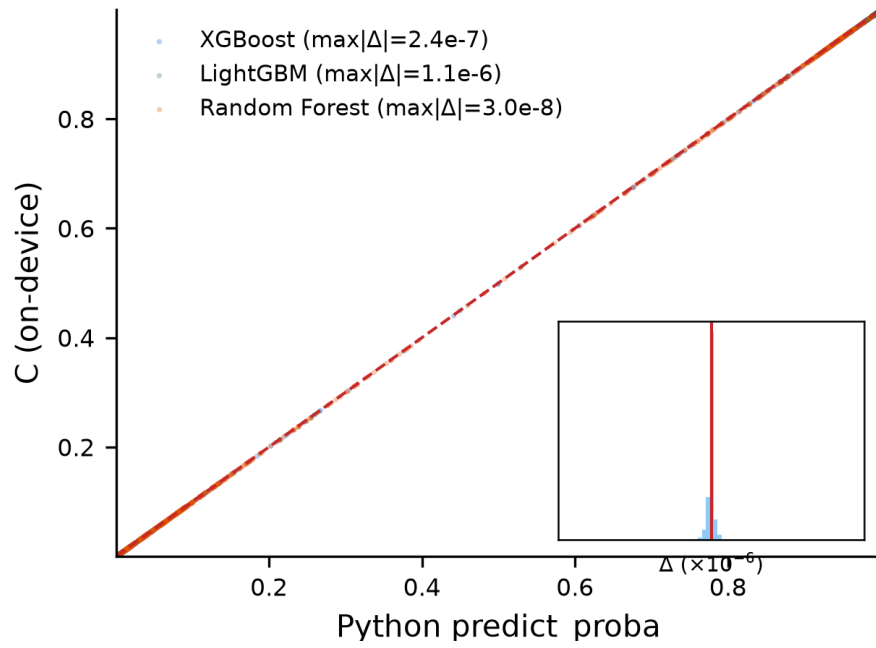


Figure 10: On-device C engine predictions versus Python `predict_proba()` over all 1 028 rows and three models (agreement scatter on the identity line, inset residual histogram of C minus Python in units of 10^{-6}). Maximum absolute errors are 2.4×10^{-7} (XGBoost), 1.1×10^{-6} (LightGBM), and 3.0×10^{-8} (Random Forest).

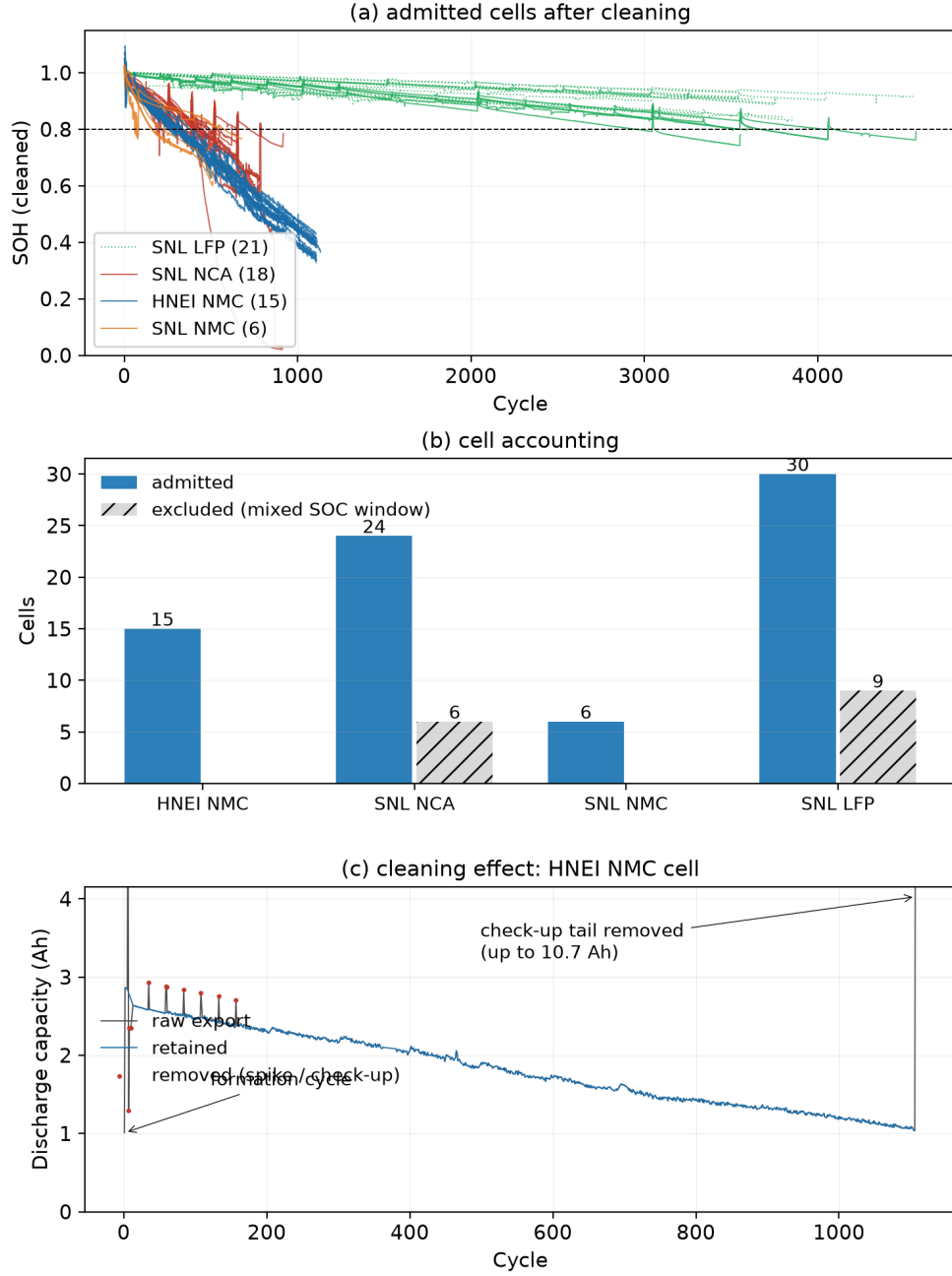


Figure 11: Data audit of the Battery Archive groups after the pre-registered cleaning rule (formation-cycle drop, Hampel despiking, level-shift truncation, voltage validity, minimum 20 usable cycles): (a) admitted-cell capacity traces with the 80% end-of-life criterion; (b) retained versus excluded cycles per group; (c) recorded exclusion reasons. Of 75 downloaded cells, 60 are admitted and 15 are excluded for exactly one recorded reason.